Subject Code: BMA 204 | Subject: Theory of Machines

Time :1 ½Hours | Maximum Marks : 30

Note:Q. 1 is compulsory.

Q1			
(a)	Chose the right answer(s): (i)Which type of kinematic pair allows both rotation and translation along a fixed axis? A) Revolute pair B) Prismatic pair C) Cylindrical pair D) Screw pair (ii)Grubler's rule provides a formula to calculate: A) Efficiency of a mechanism B) Power transmitted through a mechanism C) Degree of freedom of a mechanism D) Velocity ratio of a mechanism (iii)What is the primary purpose of kinematic analysis of planar mechanisms? A) To determine the material properties of the links B) To optimize the shape of the mechanism C) To analyze the motion and constraints of the mechanism D) To calculate the efficiency of the mechanism	(3,2,2.5,2.5) UNIT-1	CO_1
(b)	Chose the right answer(s): (i) Crank & slotted lever mechanism is an application of A) 4R B) 3P-1R C) 3R-1P D) 2R-2P (ii) Rotory engine mechanism is an application of A) 4R B) 3P-1R C) 3R-1P D) 2R-2P		
(c)	Make a neat sketch and brief about following for a radial CAM with a roller follower:(i) Base circle(ii) Trace point (iii) Pitch curve.	UNIT-2	CO_2
(d)	Differentiate between free vibrations and forced vibrations with suitable example?	UNIT-2	

Q2	(Attempt any Two Parts) UNIT-1 (CO_1)	(5,5)
	(a) With neat sketch explain the working of Whitworth Quick-Return Mechanism.	
	(b) In a four-link mechanism(shown in Fig.1), the dimensions of the links are as under: AB =50mm, BC = 66mm, CD = 56mm and AD = 100 mm At the instant when angle DAB = 60°, the link AB has an angular velocity of 10 5 rad/s in the counter-clockwise direction. Determine the (i) velocityofthepointC (ii) velocityofthepointEonthelinkBCwhen BE =40 mm	

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(iii) angular velocities of the links BC and CD

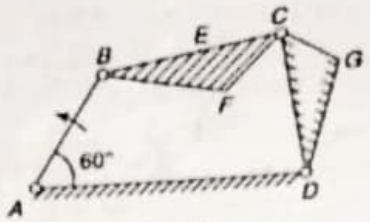


Fig. 1

(c)Find degree of freedom for the linkages shown in Fig. 2.

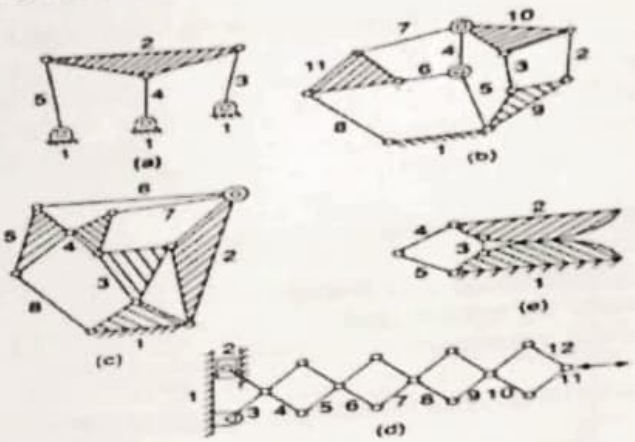


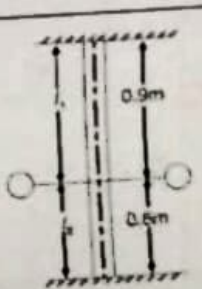
Fig.2

(5,5)

Q3 (Attempt any Two Parts) UNIT-2 (CO_2)

(a) A radial cam is to be designed for a knife edge follower with the following data: 1) Cam Lift = 50 mm during 80° of cam rotation for follower to move with simple harmonic motion; dwell for the next 40°; during the next 60° of cam rotation, the follower returns to its original position with simple harmonic motion; dwell during the remaining 180 degree. The radius of base circle of the cam is 50 mm. Write the steps for drawing the CAM and further, draw the profile of cam when the follower passes through the axis of cam shaft.

(b) A flywheel is mounted on a vertical shaft as shown in Figure. The bothends of the shaft are fixed and its diameter is 50 mm. The flywheel has a mass of 600 kg. Find thenatural frequencies of longitudinal and transverse vibrations. Take E = 200 GN/m2.



(c)With neat sketches, explain how Displacement, Velocity and Acceleration Diagrams are made for a cam and follower mechanism when the Follower moves with Simple Harmonic Motion.